# PENETRATION TESTING REPORT

## FOOTPRINTING & Reconnaissance & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

| Pentester Name (Cybersecurity Professional) | Emmanuel John |
|---|---|
| **Program/Batch** | B084 - Networkwalks |
| **Date** | 12 November 2026 |
| **Modules completed** | W2-PM1 (Multiple Kali Tools)<br>W2-PM3 (Maltego)<br>W2-PM5 (Zenmap Scanning) |
| **Client/Target** | 1. Networkwalks (secured written permission already)<br>2. My own local LAN Network |
| **Permission secured from client?** | Yes |
| **Phases covered** | **Phase 1:** Reconnaissance & Footprinting<br>**Phase 2:** Scanning & Network Discovery<br>**Phase 3-5:** In Progress |

## 1. Liability Disclaimer

I have performed these activities only on the systems and devices where I had secured written permission or the devices/systems that I own myself. All materials are for education and research purposes only. The instructor, the authors, and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job, and a permanent record.

## 2. Introduction

This report covers the footprinting of the networkwalks.com domain using multiple Kali Linux tools (W2-PM1), automated OSINT using Maltego (W2-PM3), and scanning my own local network with Zenmap (W2-PM5). These modules demonstrate how an attacker moves from gathering passive public information to mapping live hosts on a network. All commands and tools were run exclusively on my Kali Linux machine.

## 3. Tools Used

The table below lists each tool used in this report and its purpose.

| Tool | Purpose |
|---|---|
| Kali Linux & Windows | Operating systems used for reconnaissance activities |
| WHOIS | Find domain registration details (owner, dates, name servers). |
| whatweb | Fingerprint web technologies (server, CMS, plugins, IP). |
| nslookup | Resolve the domain name to its IP address using DNS. |
| curl -I | Read the HTTP response headers of the website. |
| wafw00f | Detect whether a Web Application Firewall protects the site. |
| dnsrecon | Enumerate all DNS records (NS, MX, SPF, TXT, SRV). |
| Zenmap (Nmap GUI) | Scan the local subnet to find live hosts, IPs and MAC addresses. |
| Kali linux terminal | Local IP and MAC address identification |
| Maltego | Automated OSINT graph generation and infrastructure mapping. |

## 4. Activities Performed

### 4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools.

- **WHOIS:** Identified GoDaddy as the registrar (created Nov 2019) and confirmed name servers point to ns6135.hostgator.com.
- **WhatWeb:** Identified the target runs Apache, WordPress 7.1.4, and WP Download Manager. It exposed the server IP (192.232.216.135) and the email info@networkwalks.com.
- **Nslookup:** Resolved the domain to 192.232.216.135.
- **cURL:** The headers exposed the WordPress REST API endpoint (/wp-json/) and secure cookie flags (HttpOnly, Secure).

- **Wafw00f:** Confirmed the site is protected by ModSecurity (SpiderLabs).
- **DNSRecon:** Revealed Bind version 9.16.23-RH, the Mail Server (mail.networkwalks.com), SPF records, and cPanel SRV records.

## 4.2 Automated OSINT with Maltego

To map the infrastructure visually, I used Maltego on Kali Linux.

- I created a Domain entity for networkwalks.com and ran the [Utilities] To Email Addresses [Search Engine] transform, which confirmed the info@networkwalks.com address.

- I then executed the Footprint XXL automated machine. This expanded the graph to reveal the hosting infrastructure, identifying the HostGator Name Servers (ns6136.hostgator.com) and mapping the target to specific IP netblocks (192.232.192.0-192.232.255.255 and 50.87.144.0-50.87.151.255).

## 4.3 Network Scanning with Zenmap

For the final activity, I used Zenmap on my Kali Linux machine to perform network discovery on my local network.

**Local IP:** My Kali machine is 192.168.100.106 (Subnet: 192.168.100.0/24).

**Live Hosts:** Zenmap identified 10 live hosts on the network, including the gateway router and various IoT/network devices.

**Topology:** I generated a network topology map and saved it as a PDF file to visualize the active devices.

## 5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

| # | Risk / Finding | Evidence / Observation | Potential Impact | Risk Level |
|---|---|---|---|---|
| 1 | Web technology information exposed | WhatWeb identified WordPress and WP Download Manager | Attackers may use exposed technology/version information to identify software requiring further security review | ● **Medium** |
| 2 | Server IP address identifiable | Nslookup resolved the domain to 192.232.216.135 | Provides information about the network location of the web service | ● **Low** |
| 3 | HTTP technical information exposed | Curl returned HTTP response headers and exposed /wp-json/ | May assist technology fingerprinting and further enumeration | ● **Low** |
| 4 | WAF technology identifiable | Wafw00f identified ModSecurity (SpiderLabs) | Reveals information about the web application's security architecture | ● **Low** |
| 5 | DNS infrastructure information exposed | DNSRecon identified DNS, mail and service-related records | DNS information can help build a broader infrastructure profile | ● **Medium** |
| 6 | Multiple live hosts visible on local network | Zenmap identified four live hosts in the example network | Unknown or unauthorized devices may potentially be present on a network | ● **Medium** |

**Risk level key:** ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

## 6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
   Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.

2. **Keep software updated**
   CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
   HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly**
   DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
   Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
   Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
   Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
   Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

## 7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering manual footprinting (PM1), automated OSINT (PM3), and network scanning (PM5).

By using manual tools (WHOIS, cURL, etc.) and automated tools (Maltego), I learned how much sensitive information can be gathered passively without the target ever knowing. I also learned that a single domain can expose its email addresses, hosting provider, and IP netblocks, all of which create a larger attack surface for social engineering and exploitation.

In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts, collecting IP and MAC addresses to create a network topology. The exercises showed me that information gathering is a critical part of cybersecurity, and a good report must clearly explain what was performed, what was discovered, and how to reduce the risk.

# 8. Evidences Collected

## Evidence 1: WHOIS Output

```
└─$ whois networkwalks.com
   Domain Name: NETWORKWALKS.COM
   Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.godaddy.com
   Registrar URL: http://www.godaddy.com
   Updated Date: 2025-11-12T10:08:43Z
   Creation Date: 2019-11-06T22:51:46Z
   Registry Expiry Date: 2027-11-06T22:51:46Z
   Registrar: GoDaddy.com, LLC
   Registrar IANA ID: 146
   Registrar Abuse Contact Email: abuse@godaddy.com
   Registrar Abuse Contact Phone: 480-624-2505
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Name Server: NS6135.HOSTGATOR.COM
   Name Server: NS6136.HOSTGATOR.COM
   DNSSEC: unsigned
```

## Evidence 2: WhatWeb Output

```
└─$ whatweb https://networkwalks.com
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnVar2=50511&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
```

## Evidence 3: Nslookup Output

```
└─$ nslookup networkwalks.com
Server:         192.168.100.1
Address:        192.168.100.1#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

## Evidence 4: cURL Headers Output

```
└─$ curl -I https://networkwalks.com
HTTP/1.1 200 OK
Date: Thu, 12 Nov 2026 14:10:23 GMT
Server: Apache
Permissions-Policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
Link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
Set-Cookie: __wpdm_client=99934be250b7228757d9c9cdf84c89f5; path=/; domain=networkwalks.com; secure; HttpOnly
Referrer-Policy: no-referrer-when-downgrade
X-Endurance-Cache-Level: 0
X-nginx-cache: WordPress
Content-Type: text/html; charset=UTF-8
```

## Evidence 5: Wafw00f Output

```
└─$ wafw00f networkwalks.com

                ______
               /      \
              (  W00f! )
               \  ____/
                ,,    __           404 Hack Not Found
           |`-.__   / /                      __     __
           /"  _/  /_/                       \ \   / /
          *===*    /                         \ \_/ /  405 Not Allowed
         /     )__//                         \   /
   /|  /     /---`                        403 Forbidden
   \\/`   \ |                                 / _ \
    `\    /_\\_              502 Bad Gateway  / / \ \  500 Internal Error
      `_____``-`                            /_/   \_\\

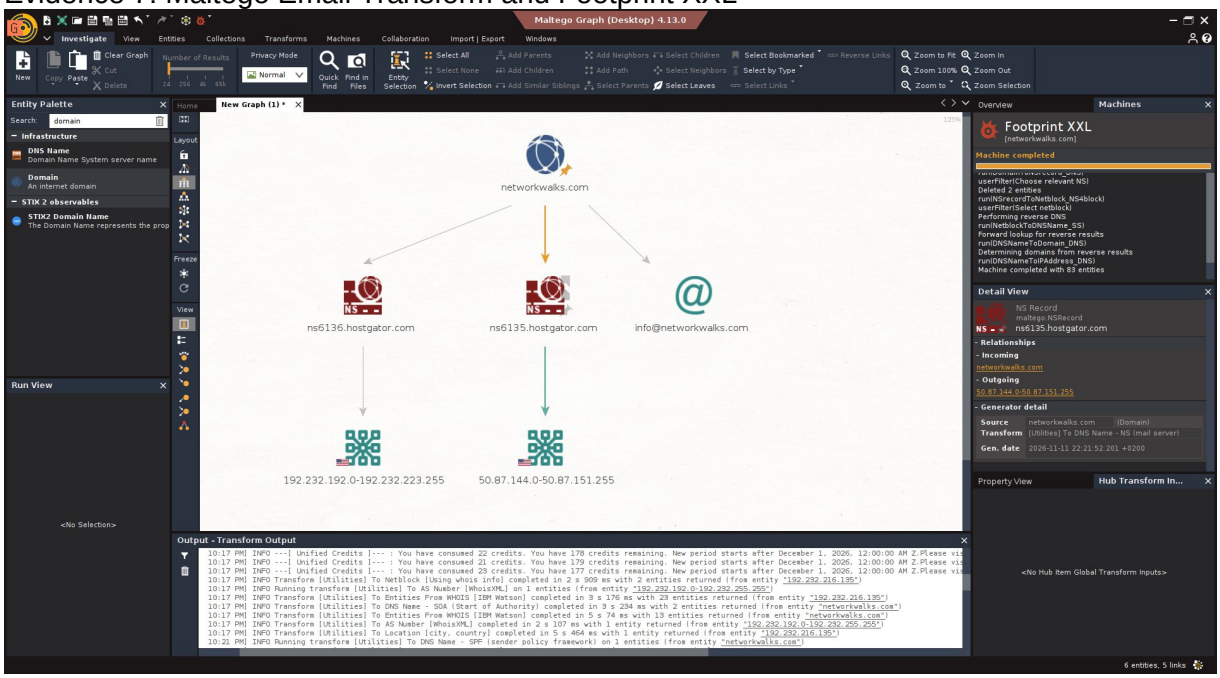
                            ~ WAFW00F : v2.4.2 ~
        The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
```

## Evidence 6: DNSRecon Output

```
└─$ dnsrecon -d networkwalks.com
2026-11-12T16:11:40.806579+0200 INFO Starting enumeration for domain: networkwalks.com
2026-11-12T16:11:40.806861+0200 INFO std: Performing General Enumeration against: networkwalks.com...
2026-11-12T16:11:43.366239+0200 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-12T16:11:43.710589+0200 INFO     SOA ns6135.hostgator.com 50.87.144.87
2026-11-12T16:11:44.400206+0200 INFO     NS ns6136.hostgator.com 192.232.216.131
2026-11-12T16:11:45.160314+0200 INFO     Bind Version for 192.232.216.131 "9.16.23-RH"
2026-11-12T16:11:45.160539+0200 INFO     NS ns6135.hostgator.com 50.87.144.87
2026-11-12T16:11:45.664776+0200 INFO     Bind Version for 50.87.144.87 "9.16.23-RH"
2026-11-12T16:11:46.482478+0200 INFO     MX mail.networkwalks.com 192.232.216.135
2026-11-12T16:11:46.490711+0200 INFO     A networkwalks.com 192.232.216.135
2026-11-12T16:11:47.403091+0200 INFO     TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-12T16:11:47.403212+0200 INFO     TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-11-12T16:11:48.017525+0200 INFO Enumerating SRV Records
2026-11-12T16:11:49.488521+0200 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.11 443
2026-11-12T16:11:49.488696+0200 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.14 443
2026-11-12T16:11:49.488794+0200 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443
2026-11-12T16:11:49.488864+0200 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
2026-11-12T16:11:49.488944+0200 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.15 443
2026-11-12T16:11:49.489021+0200 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.14 443
2026-11-12T16:11:49.489092+0200 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.15 443
2026-11-12T16:11:49.489161+0200 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.9 443
2026-11-12T16:11:49.490295+0200 INFO 8 Records Found
2026-11-12T16:11:49.490443+0200 INFO Completed enumeration for domain: networkwalks.com
```

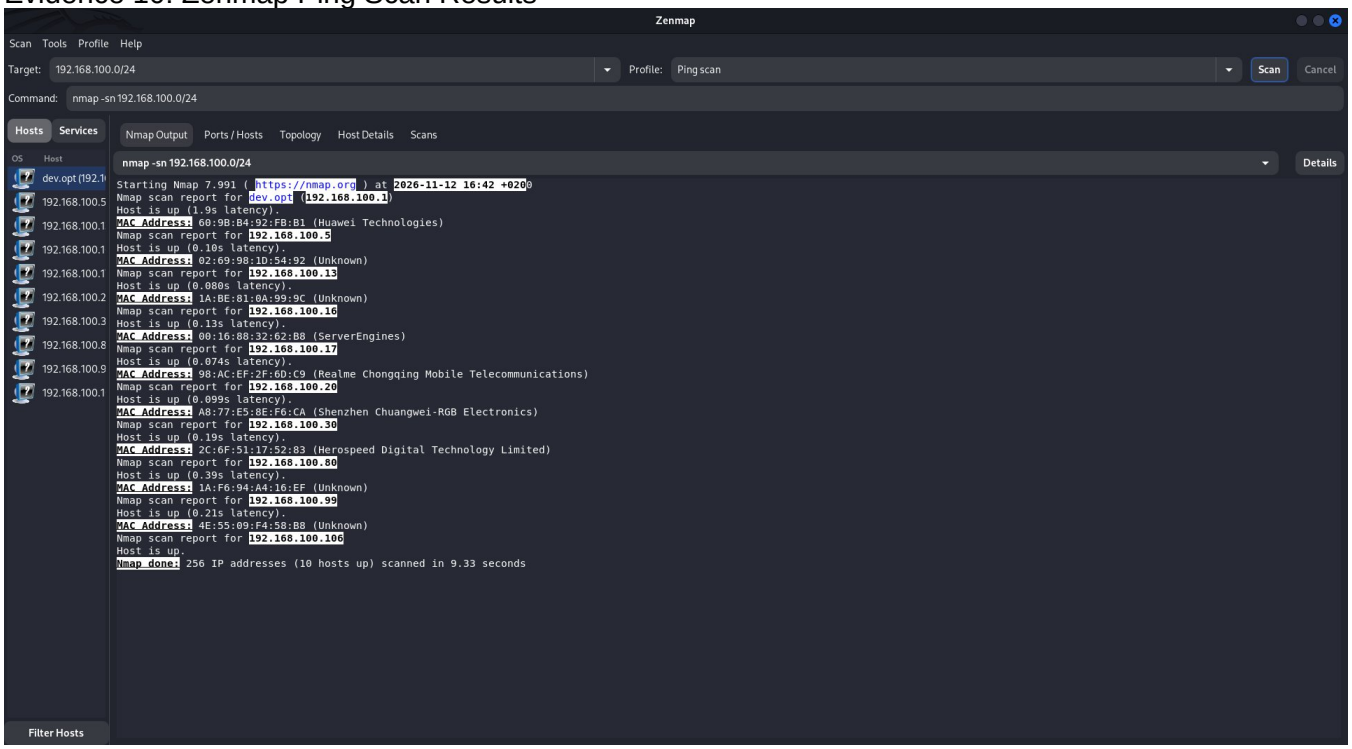
## Evidence 7: Maltego Email Transform and Footprint XXL



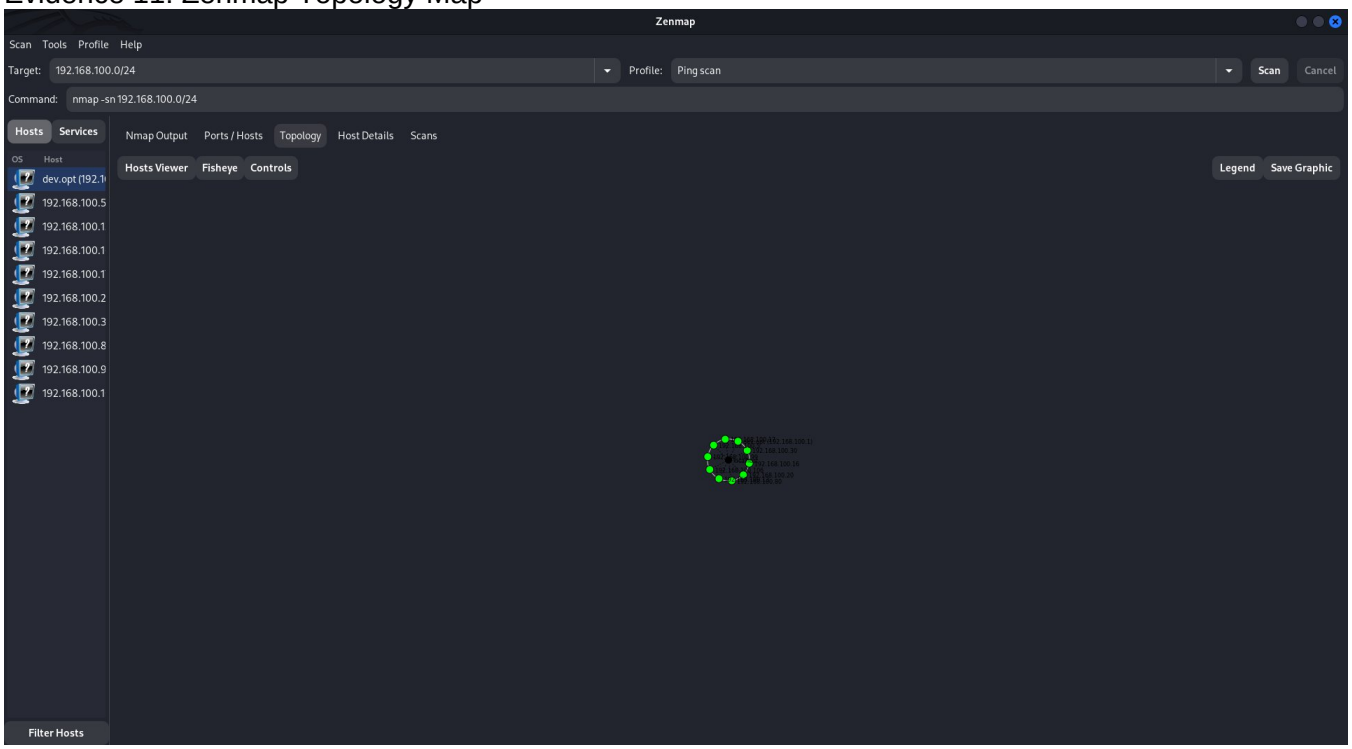

## Evidence 9: Local IP Address

```
└─$ ifconfig  wlan0
wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.100.106  netmask 255.255.255.0  broadcast 192.168.100.255
        inet6 fe80::9de1:60f:736c:37f7  prefixlen 64  scopeid 0x20<link>
        ether 34:cf:f6:0a:04:33  txqueuelen 1000  (Ethernet)
        RX packets 3573316  bytes 4032792614 (3.7 GiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1437515  bytes 410762559 (391.7 MiB)
        TX errors 0  dropped 134 overruns 0  carrier 0  collisions 0
```

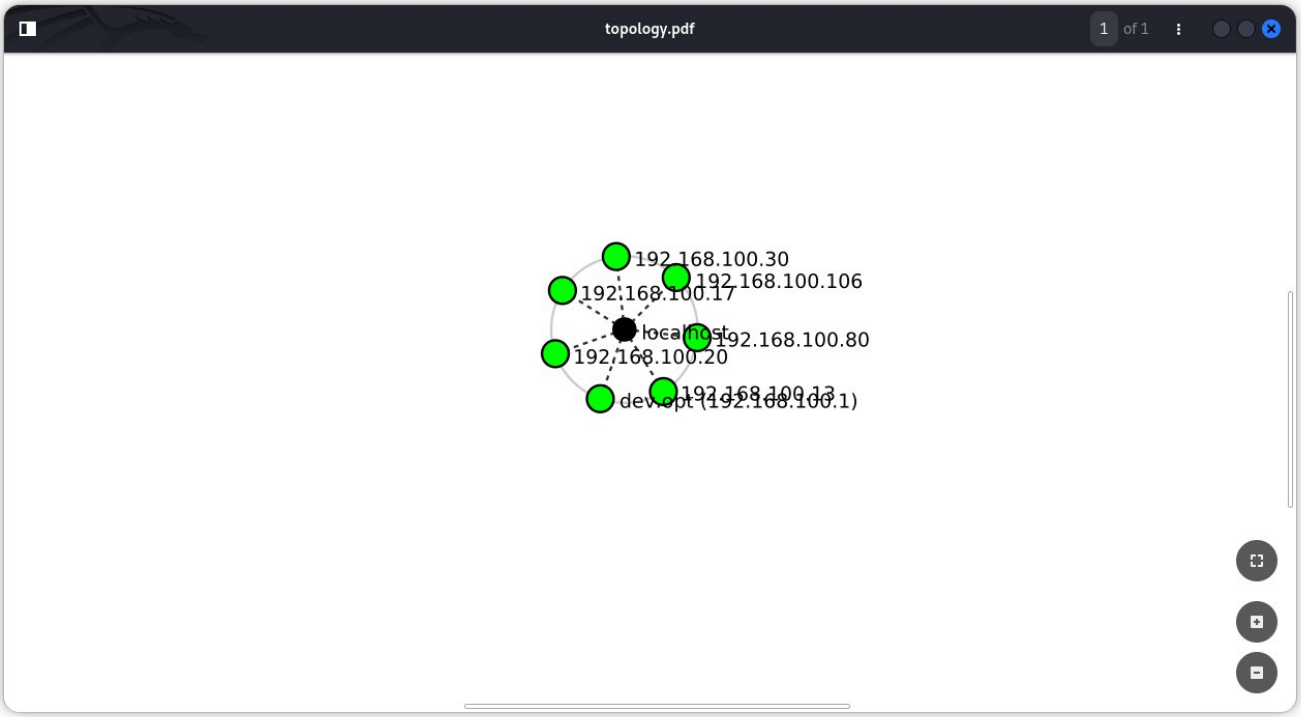
## Evidence 10: Zenmap Ping Scan Results




## Evidence 11: Zenmap Topology Map



Evidence 12: Saved Topology PDF




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**Project Information**
**Program Name:** Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** https://github.com/Ziad-1047/Networkwalks-Week2-Footprinting-B084/